Theoretical High-Efficiency Extraction Study of a Short-Pulse Electron-Beam-Pumped ArF Laser Amplifier with Atmospheric Pressure Ar-Rich Mixtures

YOUNG-WOO LEE, EIICHI MATSUI, FUMIHIKO KANNARI, AND MINORU OBARA, MEMBER, IEEE

Abstract—**Single-pass (50 cm) amplifier performance of an atmospheric-pressure ArF laser pumped by a 65 ns full width at half maximum (FWHM) short-pulse electron beam (e-beam) was investigated theoretically for a wide range of excitation rates (0.1–2.0 MW/cm^3). Atmospheric mixtures of Ne, Ar, and F_2 (three mixtures of Ar = 40, 70 percent and Ne-free) were studied.**

We have constructed a kinetic numerical model of the ArF amplifier with a Ne buffer system. A one-dimensional propagation treatment considered the gain depletion and saturated absorption spatially and temporally along the optical axis. In this model the rate constants for electron quenching of ArF* of 1.6×10^{-7}, 1.9×10^{-7}, and 2.4×10^{-7} cm^3/s were used for Ar concentration of 40, 70 percent, and Ar/F_2 mixture, respectively. With this amplifier model analysis, good agreement was obtained between theory and experiment. For the three mixtures, we calculated the extracted intensity using the optimum input intensities at each excitation rate. As a result, power efficiencies of over 10 percent were predicted at excitation rates ranging from 0.5 to 2.0 MW/cm^3.

I. INTRODUCTION

RARE gas halide lasers are high-power and high-efficiency coherent light sources in the ultraviolet (UV) region of the spectrum. Possessing the shortest wavelength (193 nm) among rare gas halide lasers, the ArF laser has the highest photon energy, so that it is capable of initiating photochemical reactions by a single photon. Various applications include photochemistry [1], [2], photolithography of very large scale integration (VLSI) devices and photochemical vapor deposition (CVD) [3]–[6], laser medicine [7], and spectroscopy. Furthermore, the inertial confinement fusion (ICF) applications [8], shorter wavelength lasers have better coupling between lasers and fuel pellets, so ArF is also a potential candidate as the energy driver for the ICF [9]–[12].

Since the first observation of an e-beam pumped ArF laser oscillation by Hoffman, Hays, and Tisone [14], fundamental studies on the e-beam excited ArF laser oscillator have been performed predominantly with Ne-buffered high-pressure (>2.5 atm) mixtures [15]–[18].

In recent work, it was confirmed that the ArF laser can be a scalable system suitable as an ICF energy driver principally because it has a smaller nonsaturable absorption than KrF which will make the gain length scaling easier. One can therefore expect the efficiency of the ArF laser to be at least as high as that of the KrF laser [9]–[13]. Mandl [12] obtained an intrinsic efficiency of 7.7 percent from an oscillator using a 2 atm mixture of Ne/Ar/F_2 pumped by an e-beam (275 keV, 50 A/cm^2, 200 ns). Oscillator performance of the ArF laser in the low-pressure (0.86 atm) regime for an Ar/F_2 mixture was reported by Suda, Obara, and Noguchi [10] and Suda *et al.* [11]. They obtained an intrinsic efficiency of 3.4 percent at an excitation rate of 2.3 MW/cm^3 using a lambda-type short-pulse e-beam (2 MeV, 80 kA, 65 ns FWHM) excitation.

On the other hand, there has been increasing interest in atmospheric pressure operation with Ar- and Kr-rich mixtures for ArF [9] and KrF [20] laser amplifiers, respectively. Generally, large-scale power amplifiers for the ICF energy drivers need large apertures. Thus, atmospheric operation of the active laser medium is suitable for reducing design constraints on large-scale laser amplifiers. Characteristics of a single-pass ArF laser power amplifier in the strongly-saturated region were investigated experimentally by Lee *et al.* [9]. They obtained a power efficiency of 15.7 percent using an atmospheric pressure Ar/F_2 mixture at an excitation rate of 0.51 MW/cm^3. The small-signal gain g_0, the nonsaturable absorption coefficient α_{ns}, and the saturation intensity I_s, were also estimated. From these results, apparently atmospheric pressure ArF laser amplifiers have high I_s (1.9–3.2 MW/cm^2) and high g_0/α (>18) values. Therefore, the ArF laser is more attractive in high-power amplification and gain length scaling than the KrF laser.

Although optical damage is an important issue for ArF laser systems, the quality of optics at 193 nm has improved rapidly in recent years and more improvements are anticipated. The laser damage threshold of dielectric coated maximum reflectors for ArF lasers has achieved damage thresholds of 0.47 GW/cm^2 at 10 pps for laser

Manuscript received December 27, 1988; revised March 14, 1989. This work was supported by the TEPCO Research Foundation.

The authors are with the Department of Electrical Engineering, Faculty of Science and Technology, Keio University, Kohoku-ku, Yokohama 223, Japan.

IEEE Log Number 8929026.

0018-9197/89/0900-0253$01.00 © 1989 IEEE

pulse length of 17 ns [76]. With any further increase in peak power in the ArF laser system, optical absorption problems may emerge. Multiphoton absorption by highly intense 193 nm radiation can, however, be minimized by selecting materials such as CaF_2 and MgF_2 with higher energy band gaps instead of fused silica.

In addition, further studies are needed under various excitation rates conditions. To our knowledge, amplifier performance of an e-beam pumped ArF laser has not yet been investigated theoretically for a wide range of excitation rates. Furthermore, there has been no report of a kinetic simulation code which would be used to explain the performance of Ne-buffered ArF amplifier system, especially in the strongly-saturated region. In this paper, we report a simulation code of an e-beam pumped Ne-buffered ArF laser power amplifier. Using this one-dimensional amplifier code analysis, we will also predict the theoretical performance of an atmospheric pressure short-pulse ArF amplifier over a wide range of excitation rates (0.1–2.0 MW/cm^3).

II. Modeling

A. Simulation Code

Most theoretical analysis for rare gas halide laser amplifiers, particularly the KrF system [21]–[23], [75], have been performed for steady-state conditions using such parameters as the small-signal gain (g_0), the nonsaturable absorption (α), the saturation intensity (I_s), the gain length (L), the aperture width (D), and the input intensity (I_{in}). The method used in these studies is valid for the case of long-pulse-length input signals. However, in the case where the input beam with short pulsewidth (< 10 ns), in particular, for angular multiplexing configurations [23], [24], it is necessary to consider gain depletion and transient absorption dynamics along the amplifier gain length spatially and temporally. Using similar techniques, Kannari and Kimura [24] obtained good agreement in a KrF amplifier model when they included these transient effects and local phenomema.

The numerical procedure used in this study was almost the same as previous KrF laser simulations [19], [74]. The calculations of the conversion of the primary relativistic electron beam to the excitation of laser media are based on the point source model. However, calculations for species such as ArF* which are saturated by laser radiation employ a one-dimensional treatment along the optical axis. All of the reaction processes and their rate constants used in the present model for mixtures of $Ne/Ar/F_2$ and Ar/F_2 are listed in Table I. The number densities of rare gas ions (Ne^+ and Ar^+) were calculated initially by dividing the e-beam deposition energy by the W-values (required energy for excitation or ionization of single atom) of 36.6 eV/ion for Ne [26] and of 26.4 eV/ion for Ar [26], in which the production ratio of the rare gas ions to metastable atoms in both Ar and Ne is assumed to be 3.3 [25].

The electron reactions have rate constants which are a function of electron temperature (T_e), as shown in Table I. The mean electron temperature is calculated from the secondary electron number density (number/cm^3) calculated by the rate equations including the electron dissociative attachment rate constant and electron–ion recombination rate constant, which are strongly dependent on the electron energy distribution ($eV^{-1}cm^{-3}$). We assumed that the electron energy distribution is Maxwellian for calculating many rates such as electron-neutral elastic and inelastic collisions and electron–ion recombination as shown in Table I. However, Elliott and Greene [68] have shown that the electron energy distributions in e-beam-generated Xe and Ar plasma calculated using a Boltzmann equation were different from Maxwellian distributions. This uncertainty makes it difficult to obtain the exact predictions of the electron rate constants, such as F_2 dissociative attachment and electron collisional quenching. Rozenberg, Lando, and Rokni [69] have experimentally reported the secondary-electron density in the Ar/F_2 mixture pumped by e-beams (350 ns, 150 keV). They showed the F_2 dissociative attachment rate has an important effect on the electron distribution. Kannari and Kimura [70] have also performed numerical calculations for $Ne/Xe/HCl$ mixture pumped by e-beams in order to obtain the exact low-energy electron distributions in the e-beam-pumped XeCl laser mixture. On the other hand, Trainor and Jacob [57] have measured the electron quenching rate for the upper laser level of ArF* and reported it to be 1.6×10^{-7} cm^3/s. The electron quenching rate constant of 1.6×10^{-7} cm^3/s has been used in the KrF laser model for $Ar/Kr/F_2$ mixtures [19], but Mandl [12] used a rate constant of 2.0×10^{-7} cm^3/s obtained by using the electron attachment rate for F_2 reported in [71]. As a result of applying the rate constant of 1.6×10^{-7} cm^3/s to our model, the case of a Ar = 40 percent mixture showed good agreement with the previous ArF laser amplifier experiment [9]. The experimental results in [57] for the electron quenching rates, however, were obtained by using a theoretical rate constant such as the electron dissociative attachment to F_2 as a function of electron energy. Furthermore, Kushner and Moratz [72] showed the uncertainty in the process of a F_2 destruction mechanism in e-beam pumped excimer lasers. They reported that the process of direct electron impact dissociation of F_2 is competitive with the dissociative attachment reaction. Conversely, the electron energy distribution in rare gas dimer and rare gas halide laser mixtures pumped by e-beams is dependent upon the gas mixture [69], [70], and furthermore, the rate constant for electron quenching of KrF and XeF is also weakly dependent on the electron energy [27]. Therefore, at present it is not necessarily valid that one value of the electron rate constant applies to the three different mixtures studied here, in which the Ar concentrations vary largely from 40 to over 99 percent. Hence, it is assumed here that the electron quenching rate constants for various mixtures are slightly different because the electron energy distribution in each mixture would be different. However, although the electron quenching rate constant is not presented as a function of electron energy in the present model, it is necessary to

TABLE I
RATE CONSTANTS FOR THE KINETIC REACTIONS IN $Ne/Ar/F_2$ SYSTEMS. *Te*, *TG*, AND GAS DENSITY IN AMAGAT, RESPECTIVELY. *M* INDICATES RARE GAS ATOM

Reaction	Rate Constant	Reference
Electron Reaction		
$Ne + e_p \rightarrow Ne^+ + e_p + e$	36.6 eV/ion-electron pair	[26]
$Ne + e_p \rightarrow Ne^* + e_p$	120 eV/excited atom	[25]
$Ar + e_p \rightarrow Ar^+ + e_p + e$	26.4 eV/ion-electron pair	[26]
$Ar + e_p \rightarrow Ar^* + e_p$	93 eV/excited atom	[25]
$Ne + e \leftrightarrow Ne^* + e$	$2.55 \times 10^{-9}\, Te^{1.69} \exp(-24.9/Te)$ cm^3/s[a]	[28]
$Ne + e \leftrightarrow Ne^{**} + e$	$4.82 \times 10^{-10}\, Te^{0.48} \exp(-27.8/Te)$ cm^3/s[a]	[28]
$Ne^* + e \leftrightarrow Ne^{**} + e$	$3.83 \times 10^{-7}\, Te^{0.32} \exp(-2.85/Te)$ cm^3/s[a]	[28]
$Ne + e \rightarrow Ne^+ + 2e$	$1.23 \times 10^{-9}\, Te^{0.72} \exp(-32.4/Te)$ cm^3/s	[28]
$Ne^* + e \rightarrow Ne^+ + 2e$	$3.04 \times 10^{-8}\, Te^{0.74} \exp(-7.50/\mathrm{Te})$ cm^3/s	[28]
$Ne^{**} + e \rightarrow Ne^+ + 2e$	$1.00 \times 10^{-7}\, Te^{0.60} \exp(-4.65/Te)$ cm^3/s	[28]
$Ne_2^* + e \rightarrow Ne^+ + 2e$	$4.93 \times 10^{-6}\, Te^{0.68} \exp(-6.45/Te)$ cm^3/s	[28]
$Ne_2^* + e \rightarrow 2Ne + e$	1.0×10^{-7} cm^3/s	Guessed
$Ar + e \leftrightarrow Ar^* + e$	$3.70 \times 10^{-9}\, Te^{0.74} \exp(-17.4/Te)$ cm^3/s[a]	[29]
$AR + e \leftrightarrow Ar^{**} + e$	$1.05 \times 10^{-8}\, Te^{0.71} \exp(-19.7/Te)$ cm^3/s[a]	[29]
$Ar^* + e \leftrightarrow Ar^{**} + e$	$7.24 \times 10^{-7}\, Te^{0.51} \exp(-2.28/Te)$ cm^3/s[a]	[29]
$Ar + e \rightarrow Ar^* + 2e$	$1.75 \times 10^{-8}\, Te^{0.68} \exp(-23.6/Te)$ cm^3/s	[29]
$Ar^* + e \rightarrow Ar^+ + 2e$	$5.18 \times 10^{-6}\, Te^{0.67} \exp(-6.27/Te)$ cm^3/s	[29]
$Ar^{**} + e \rightarrow Ar^+ + 2e$	$1.41 \times 10^{-7}\, Te^{0.61} \exp(-4.00/Te)$ cm^3/s	[29]
$AR_2^* + e \rightarrow Ar^+ + 2e$	$6.78 \times 10^{-8}\, Te^{0.70} \exp(-5.25/Te)$ cm^3/s	[29]
$Ne_2^* + e \rightarrow 2Ne + e$	1.0×10^{-7} cm^3/s	Guessed
$F_2 + e \rightarrow F^- + F$	$4.0 \times 10^{-9}\, Te^{-1.16}$ cm^3/s	[30]
$F_2 + e \rightarrow 2F + e$	3.0×10^{-10} cm^3/s	[31]
$F + e \rightarrow F^-$	1.0×10^{-12} cm^3/s	[31]
Neutral Reaction		
$Ne^* + 2Ne \rightarrow Ne_2^* + Ne$	4.1×10^{-34} cm^6/s	[32]
$Ar^* + 2Ar \rightarrow Ar_2^* + Ar$	1.1×10^{-32} cm^6/s	[33]
$Ar^* + Ar + Ne \rightarrow Ar_2^* + Ne$	1.1×10^{-32} cm^6/s	[34]
$Ne^{**} + Ne \rightarrow Ne^* + Ne$	7.1×10^{-11} cm^3/s	[35]
$Ar^{**} + Ar \rightarrow Ar^* + Ar$	1.0×10^{-10} cm^3/s	[36]
$Ar^{**} + Ne \rightarrow Ar^* + Ne$	1.0×10^{-10} cm^3/s	[37]
$F^* + Ar \rightarrow Ar^* + F$	1.0×10^{-10} cm^3/s	Guessed
Penning Ionization		
$Ne^* + Ar \rightarrow Ar^+ + Ne + e$	6.5×0^{-11} cm^3/s	[38]
$Ne^* + Ar \rightarrow NeAr^+ + e$	2.3×10^{-11} cm^3/s	[38]
$Ne_2^* + Ar \rightarrow Ar^+ + Ne + e$	6.5×10^{-11} cm^3/s	[38]
$Ne_2^* + Ar \rightarrow NeAr^+ + Ne + e$	2.3×10^{-1} cm^3/s	[38]

Reaction	Rate Constant	Reference
Charge Transfer		
$Ne^+ + 2Ne \rightarrow Ne_2^+ + Ne$	4.4×10^{-32} cm^6/s	[39]
$Ne^+ + Ar + Ne \rightarrow NeAr^+ + Ne$	1.0×10^{-31} cm^6/s	[40]
$Ne^+ + 2Ar \rightarrow NeAr^+ + Ar$	1.0×10^{-31} cm^6/s	[41]
$Ar^+ + 2Ar \rightarrow Ar_2^+ + Ar$	2.5×10^{-31} cm^6/s	[41]
$Ar^+ + Ar + Ne \rightarrow Ar_2^+ + Ne$	3.0×10^{-31} cm^6/s	[42]
$Ar^+ + 2Ne \rightarrow NeAr^+ + Ne$	$1.0\ 10^{-31}$ cm^6/s	[40]
$Ne_2^+ + Ar + Ne \rightarrow Ar^+ + 3Ne$	3.5×10^{-30} cm^6/s	[43]
$Ne_2^+ + 2Ar \rightarrow Ar^+ + Ar + 2Ne$	3.5×10^{-31} cm^6/s	[43]
$Ne_2^+ + Ar \rightarrow Ar^+ + Ne$	3.0×10^{-11} cm^3/s	[43]
$NeAr^+ + Ar \rightarrow Ar_2^+ + Ne$	1.2×10^{-11} cm^3/s	[44]
$NeAr^+ + Ar \rightarrow Ar^+ + Ne + Ar$	1.1×10^{-9} cm^3/s	[44]
Electron-Ion Recombination		
$Ne_2^+ + e \rightarrow Ne^{**} + Ne$	$3.2 \times 10^{-9}\, (Tg/Te)^{0.43}$ cm^3/s	[45]
$Ar_2^+ + e \rightarrow Ar^{**} + Ar$	$1.5 \times 10^{-9}\, (Tg/Te)^{0.67}$ cm^3/s	[45]
$NeAr_2^+ + e \rightarrow Ar^{**} + Ne$	$1.5 \times 10^{-9}\, (Tg/Te)^{0.67}$ cm^3/s	[46]
Radiation		
$Ne_2^* \rightarrow 2Ne + h\nu$	8.9×10^{-7} s^{-1}	[47]
$Ar_2^* \rightarrow 2Ar + h\nu$	6.0×10^{-7} s^{-1}	[48]
Ion-Ion Recombination		
$Ar^+ + F^- + Ar \rightarrow ArF^* + Ar$	$\frac{9.32 \times 10^{-6} N}{N^2 + 6.21 \times 10^{-2} N + 1.69}$ cm^3/s[b]	[49]
$Ar^+ + F^- + Ne \rightarrow ArF^* + Ne$	$\frac{4.59 \times 10^{-5} N}{N^2 + 3.93 N + 19.5}$ cm^3/s[b]	[49]
$Ar_2^+ + F^- + Ar \rightarrow ArF^* + 2Ar$	$\frac{1.51 \times 10^{-5} N}{N^2 + 1.81 N + 2.25}$ cm^3/s[b]	[49]
$Ar_2^+ + F^- + Ne \rightarrow ArF^* + Ar + Ne$	$\frac{3.94 \times 10^{-5} N}{N^2 + 0.589 N + 18.2}$ cm^3/s[b]	[49]
$Ne^+ + F^- + Ar \rightarrow NeF^* + Ar$	$\frac{1.39 \times 10^{-5} N}{N^2 + 0.919 N + 1.38}$ cm^3/s[b]	[49]
$Ne^+ + F^- + Ne \rightarrow NeF^* + Ne$	$\frac{2.80 \times 10^{-5} N}{N^2 + 1.44 N + 10.9}$ cm^3/s[b]	[49]
$Ne_2^+ + F^- + Ar \rightarrow NeF^* + Ne + Ar$	$\frac{1.16 + 10^{-5} N}{N^2 + 0.475 N + 1.56}$ cm^3/s[b]	[49]
$Ne_2^+ + F^- + Ne \rightarrow NeF^* + 2Ne$	$\frac{3.66 \times 10^{-5} N}{N^2 + 1.74 N + 17.3}$ cm^3/s[b]	[49]
$NeAr^+ + F^- + Ar \rightarrow ArF^* + Ne + Ar$	$\frac{1.15 + 10^{-5} N}{N^2 + 0.490 N + 1.95}$ cm^3/s[b]	[49]
$NeAr^+ + F^+ + Ne \rightarrow ArF^* + 2Ne$	$\frac{5.27 \times 10^{-5} N}{N^2 + 6.81 N + 20.3}$ cm^3/s[b]	[49]

[a]Rate constants of inelastic collision process are included. Superelastic collision processes are treated by the principle of a detailed balance.

TABLE I
(*Continued*)

Reaction	Rate Constant	Reference
Neutral Formation Reaction		
$Ar^* + F_2 \rightarrow ArF^* + F$	7.5×10^{-10} cm^3/s	[50]
$Ar_2^* + F_2 \rightarrow ArF^* + Ar + F$	5.2×10^{-10} cm^3/s	[51]
$Ar_2^* + F_2 \rightarrow Ar_2F^* + F$	2.5×10^{-10} cm^3/s	[52]
$Ar^{**} + F_2 \rightarrow ArF^* + F$	7.5×10^{-10} cm^3/s	[53]
$Ar + F_2^* \rightarrow ArF^* + F$	7.5×10^{-10} cm^3/s	[53]
$Ne^* + F_2 \rightarrow NeF^* + F$	5.3×10^{-10} cm^3/s	[32]
$Ne^{**} + F_2 \rightarrow NeF^* + F$	7.5×10^{-10} cm^3/s	[54]
ArF Relaxation*		
$ArF^* + Ne \rightarrow Ar + F + Ne$	1.6×10^{-12} cm^3/s[c]	[55]
$ArF^* + Ar \rightarrow Ar + F + Ar$	8.6×10^{-12} cm^3/s[c]	[56]
$ArF^* + F_2 \rightarrow Ar + 3F$	1.8×10^{-9} cm^3/s[c]	[56]
$ArF^* + e \rightarrow Ar + F + e$	(See Text)	[57]
$ArF^* + 2Ar \rightarrow Ar_2F^* + Ar$	5.0×10^{-31} cm^6/s[b,c]	[58]
$ArF^* + 2Ne \rightarrow$ products	1.0×10^{-32} cm^6/s[b,c]	[55]
$ArF^* + Ar + Ne \rightarrow Ar_2F^* + Ne$	1.2×10^{-31} cm^6/s[b,c]	[59]
$ArF^*(B) \rightarrow Ar + F + h\nu$	2.38×10^{-8} s^{-1}	[58]
$ArF^*(C) \rightarrow Ar + F + h\nu$	2.08×10^{-7} s^{-1}	[58]
$ArF^*(B) + \phi \rightarrow Ar + F + 2\phi$	2.75×10^{-16} cm^2	[50]
Ar_2F^* Relaxation		
$Ar_2R^* + Ar \rightarrow 2Ar + F$	2.2×10^{-14} cm^3/s	[58]
$Ar_2F^* + F_2 \rightarrow 2Ar + 3F$	2.0×10^{-10} cm^3/s	[61]
$Ar_2F^* + e \rightarrow 2Ar + F + e$	1.0×10^{-7} cm^3/s	Guessed
$Ar_2F^* \rightarrow 2Ar + F + e$	4.4×10^{6} s^{-1}	[58]
NeF-Related Reaction*		
$NeF^* \rightarrow Ne + F^*$	2/3 of formed NeF*	[62]
$NeF^* \rightarrow Ne + F + h\nu$ $NeF^* + M \rightarrow Ne + F + M$	1/3 of formed NeF*	[62]
$F^* + F_2 \rightarrow F_2^* + F$	5.1×10^{-10} cm^3/s	[32]
$F_2^* + F_2 \rightarrow$ products	3.5×10^{-10} cm^3/s	[32]
$F_2^* \rightarrow F_2 + h\nu$	2.7×10^{8} s^{-1}	[27]
Absorption		
$Ne_2^+ + \phi \rightarrow Ne^+ + Ne + e$	2.3×10^{-19} cm^2	[63]
$Ne_2^* + \phi \rightarrow Ne_2^+ + e$	9.2×10^{-19} cm^2	[64]
$Ne^{**} + \phi \rightarrow Ne^+ + e$	2.5×10^{-19} cm^2	[65]
$Ne^* + \phi \rightarrow Ne^+ + e$	1.3×10^{-19} cm^2	[65]
$Ar_2^+ + \phi \rightarrow Ar^+ + Ar + e$	1.0×10^{-19} cm^2	Guessed
$Ar_2^* + \phi \rightarrow Ar_2^+ + e$	6.3×10^{-19} cm^2	[64]
$Ar^{**} + \phi \rightarrow Ar^+ + e$	3.0×10^{-18} cm^2	[65]
$Ar^* + \phi \rightarrow Ar^+ + e$	4.0×10^{-20} cm^2	[65]
$F^- + \phi \rightarrow F + e$	1.0×10^{-18} cm^2	[66]
$F_2 + \phi \rightarrow 2F$	1.5×10^{-21} cm^2	[57]
$Ar_2F^* + \phi \rightarrow 2Ar + F$	5.0×10^{-19} cm^2	Guessed

[b]At a gas temperature of 300 K.
[c]Based on the radiative lifetime of ArF*(B).

modify this rate constant for each mixture. Then, we used three different rate constants for the electron quenching process which were obtained by the best laser parameter fitting to the experimental results [9]. In this study, the electron quenching rate constants used are 1.6×10^{-7}, 1.9×10^{-7}, and 2.4×10^{-7} cm^3/s for Ar concentrations of 40, 70 percent, and the Ar/F_2 mixture, respectively.

The calculation of the laser beam propagation in the amplifier employs a one-dimensional treatment along the optical axis. The gain-length (L = 50 cm) based on the experiment [9] of amplifier is divided into 20 segments of considering spatial variation of gain and absorption. In each zone of length (= 2.5 cm) the number densities of all species are assumed to be the same during the beam propagation in each zone. The propagation beam intensity $I(x, t)$ is amplified along the amplifier length from $x = 0$ to $x = L (L = 50$ cm), which is described by

$$\frac{1}{c}\frac{\partial I(x,t)}{\partial t} + \frac{\partial I(x,t)}{\partial x} = I(x,t)\{g(x,t) - \alpha(x,t)\} \quad (1)$$

where c is the speed of light, $g(x, t)$ is the gain coefficient, and $\alpha(x, t)$ is the absorption coefficient. We assumed that the beam intensity in the amplifier does not vary with time during each step of calculation of the rate equations. Equation (1) can be reduced to (2) at each time step, and the input beam intensity $I(0)$ is amplified along the amplifier region from $x = 0$ to $x = L (L = 50$ cm) at each time step by the following equation.

$$\frac{\partial I(x)}{\partial x} = I(x)\{g(x) - \alpha(x)\} \quad (2)$$

$$g(x) = \sigma_{st} N^*(x) \quad (3)$$

where σ_{st} is the effective cross section for stimulated emission and $N^*(x)$ is the number density of the ArF(B). We assumed the gain shape as function of frequency is Gaussian near the line center having a bandwidth ($\Delta\lambda$) of 15 Å [60]. $\sigma_{st} = 2.75 \times 10^{-16}$ cm^2 was used in this calculation.

The absorption coefficient $\alpha(x)$ is written as

$$\alpha(x) = \sum_i \sigma_{\alpha i} N_{\alpha i}(x) \quad (4)$$

where $N_{\alpha i}(x)$ is the number density of species i which absorb the laser photons and $\sigma_{\alpha i}$ is the absorption cross section of the respective species.

The effect of the amplified spontaneous emission (ASE) is a quite important and often a limiting factor for design-

ing large-scale amplifiers [21], [22]. However, we have ignored the effect of ASE because we have taken an intense input intensity corresponding to strongly-saturated conditions which can greatly reduce the ASE effect. Furthermore, the previous experiment [9] has shown that the effect of ASE in the ArF laser amplifier was much smaller than that of the KrF laser amplifier experiment [20], even at the small-signal amplification region.

Fig. 1 shows a schematic diagram explaining the theoretical model of the ArF laser amplifier described above. In this study, we do not consider the effect of the unpumped region and the windows of the laser amplifier. We will investigate only the "single-pass" amplification characteristics of the ArF laser amplifier having a 50 cm gain length, which is advantageous to clarify the fundamental amplifier properties rather than the multipass amplifier.

Since many rate constants are functions of gas temperature (T_g), as for example, the three-body quenching and the ion–ion recombination reactions, the present kinetics code calculates the gas temperature as a function of time, and used kinetic rates as a function of the instantaneous gas temperature. Fig. 2 shows the time dependence of the gas temperature for an Ar/F_2 mixture at excitation rates of 0.5, 1.0, and 2.0 MW/cm^3.

The case for a high excitation rate of 2 MW/cm^2 shows a significant temperature rise at the end of e-beam current pulse.

B. Code Verification

Fig. 3 shows typical time histories of the e-beam current, the sidelight fluorescence, the input laser pulse, and the output laser pulse obtained in the previous experiment [9]. The input laser beam shape used in this study was assumed to be a Gaussian pulse having a pulsewidth of 20 ns FWHM, which was very similar to the pulse shape in the experimental measurement. We evaluated the input–output performance of the amplifier at the peak of the input and output laser pulses.

Fig. 4 shows the calculated time histories of the small-signal gain g_0, the loaded gain g, the absorption coefficient α, the input beam, the output beam, and the excitation rate under the same conditions as the case of Ar = 70 percent mixture in the previous experiment [9]. The g shows the depleted gain curve obtained at the amplifier exit (L = 50 cm) when the input beam with a pulsewidth of 20 ns FWHM is loaded. The excitation rate was calculated from the electron beam current pulse in Fig. 3. We always used this excitation rate pulse shape in the simulation code, and we changed the excitation rate by means of changing the peak of e-beam current density with a constant e-beam pulse width of 65 ns FWHM. In the experiment [9], the signal beam pulse was delayed into the amplifier by 30 ns from the time at which the peak of sidelight signal appeared, as shown in Fig. 3. Then, in Fig. 4, we injected the input beam at the same time as the experiment [9], and the excitation rate was also determined by considering the time delay.

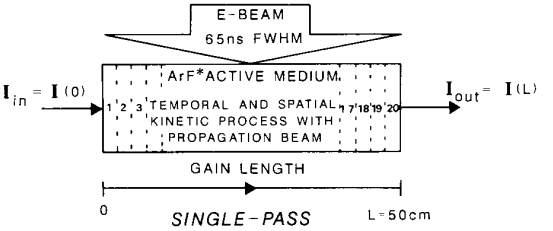


Fig. 1. Schematic representation of the ArF laser amplifier used to calculate the single-pass (L = 50 cm) amplification characteristics.

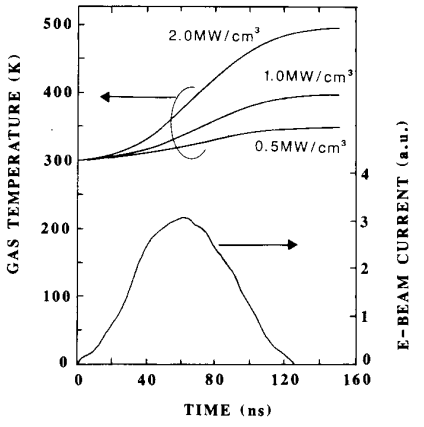


Fig. 2. Temporal variations of calculated gas temperatures of the Ar/F_2 mixture for the excitation rates of 0.5, 1.0, and 2.0 MW/cm^3. Typical time history of e-beam current pulse obtained experimentally is also shown.

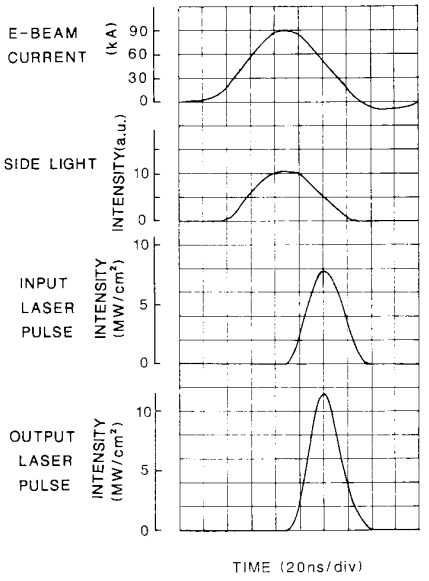


Fig. 3. Typical time histories of the e-beam current, the sidelight fluorescence, the input laser pulse, and the output laser pulse obtained in the ArF laser amplifier experiment [9]. The laser mixture was 1 atm of Ar/F_2 = 99.6/0.4 (percent).

Fig. 5(a) shows g_0, α, and Is, and (b) shows the power efficiency (solid line) together with the experimental results [9] as a function of Ar concentration. The dashed lines in Fig. 5(a) show the results using the electron

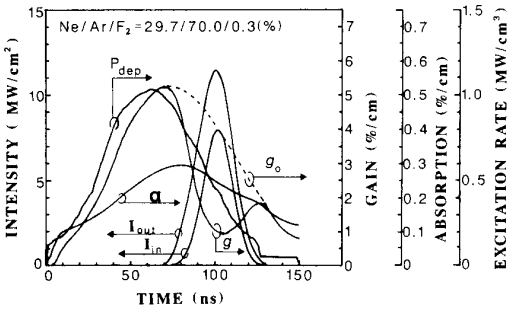


Fig. 4. Calculated time histories of the small-signal gain (g_0), the loaded gain (g), the absorption coefficient (α), and the input and output beam pulses under the same condition as the previous experiment [9] for the 1 atm gas mixture of $Ar/Ne/F_2$ = 70/29.7/0.3 (percent). Calculated excitation rate (P_{dep}) is also shown. The injection time of the input signal is considered 30 ns time delay. (See [9].)

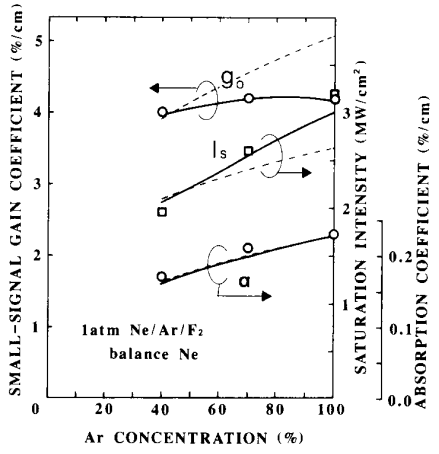


Fig. 5. (a) Plots of experimentally-estimated small-signal gain (g_0), absorption coefficient (α), and saturation intensity (I_s) as a function of Ar concentration for a 1 atm mixture of $Ar/Ne/F_2$. Theoretical curves calculated with the modified electron quenching rate constants and a constant 1.6×10 cm^3/s for all mixtures are shown by solid lines and broken lines, respectively. (b) The calculated power efficiency (solid line) as a function of Ar concentration using the present model is shown together with the experimental results (open circles).

quenching rate constant of 1.6×10^{-7} cm^3/s for all mixtures. Note that in this figure, the excitation rate increases as the Ar concentration is increased due to the constant e-beam current density. These simulation results (shown by solid lines) obtained by the same procedure and by using modified rate constants for electron quenching of ArF* described previously are in quite good agreement with the experimental results plotted in Fig. 5(a). Fig. 5(b) also shows good agreement between previous experimental results [9] and calculation using the present model.

III. Results and Discussion

A. Simulation Condition

Here, we will investigate only the single-pass (L = 50 cm) amplifier performance of the e-beam pumped atmospheric pressure ArF laser amplifier. We employed in this study the three mixtures used in the previous experiment [9], namely 60 percent Ne, 30 percent Ne, and Ne-free (Ar/F_2) mixtures.

Each F_2 concentration in the three mixtures was optimized individually to maximize the extraction efficiency for a wide range of excitation rates (0.1–2.0 MW/cm^3). Fig. 6 shows the optimum F_2 concentrations and optimum input intensities for all mixtures studied here as a function of excitation rate. The optimum input intensities for each mixture were also optimized individually to get the maximum extraction efficiency. The Ar-rich mixtures gave higher optimum input intensities, indicating that the Ar-rich mixture has larger saturation intensities than does the Ar-lean mixture. At the same excitation rate, the optimum F_2 concentration is constant, independent of Ar concentration ranging from 40 to 99 percent. Temporal variations of the small-signal gain g_0 and the absorption α are very large under the conditions of a short pulse (65 ns FWHM) pumping with a high excitation rate (2 MW/cm^3). Thus, in our model it is assumed that an input Gaussian laser pulse is injected so that its peak corresponds to the peak of the amplifier gain curve.

Fig. 7 shows typical characteristics of the input and output intensities of the ArF laser pulse at an excitation rate of 1 MW/cm^3 and the temporal property of gain depletion obtained at the amplifier exit (L = 50 cm). At this excitation rate, the peak of small-signal gain g_0 decreases with increasing Ar concentration, but the extraction intensities (the peak value of the output density I_{out} minus the peak value of the input intensity I_{in}) are similar to the three mixtures. The optimum input intensity decreases with increasing Ne concentration, but the Ar-rich mixtures (Ar = 99.4, 70 percent) have larger output intensities than that of an Ar-lean mixture (Ar = 40 percent).

B. Small-Signal Gain, Absorption, and Saturation Intensity

Fig. 8 shows the calculated small-signal gain for the three mixtures as a function of excitation rate. These values increase almost linearly with increasing excitation rates. At the same excitation rate, the small-signal gain increases with Ne concentration. In order to explain this result, we investigated the formation processes and deactivation processes of the upper laser level ArF*(B). Fig. 9 shows the fractional formation channels as a function of Ar concentration when the excitation rate is (a) 0.1 and (b) 2.0 MW/cm^3, respectively. The ArF*(B) number density increases with an increase of the Ar concentra-

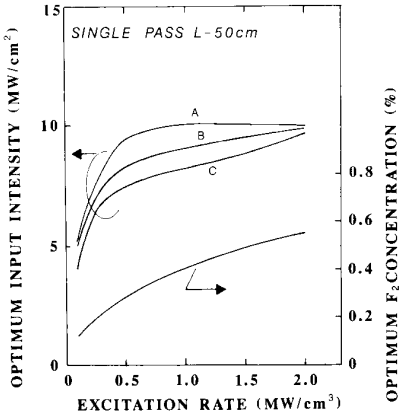


Fig. 6. Optimum F_2 concentrations and input intensities for all atmospheric mixtures studied here as a function of excitation rate. These values were optimized in each case to maximize the extraction efficiency. The designation of the symbols attached to curves is as follows. Curve *A*: $Ar/F_2 = 100 - x/x$ (percent), curve *B*: $Ar/Ne/F_2 = 70/30 - x/x$ (percent), curve *C*: $Ar/Ne/F_2 = 40/60 - x/x$ (percent), where x is the optimum F_2 concentration as shown in this figure.

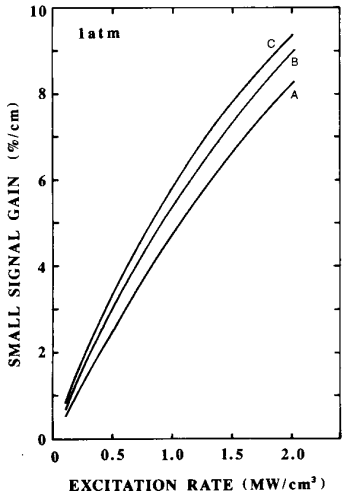


Fig. 8. Calculated small-signal gain as a function of excitation rate for each optimum mixture. The designation of the curves is the same as that of Fig. 6.

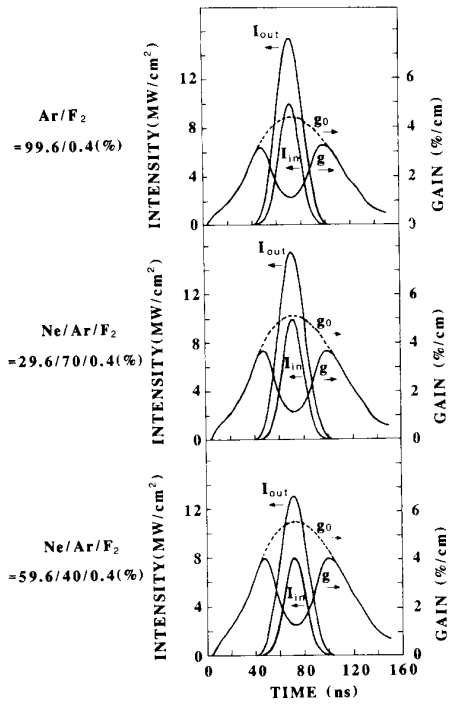


Fig. 7. Typical characteristics of the output beam pulses calculated by the beam transport equation in the present amplifier model together with corresponding input beam pulses for the three gas mixtures (Ar/F_2, Ar = 70, and Ar = 40 percent) pumped at a excitation rate of 1 MW/cm³. Small-signal gain (g_0) and the loaded gain (g) at the amplifier exit are also shown.

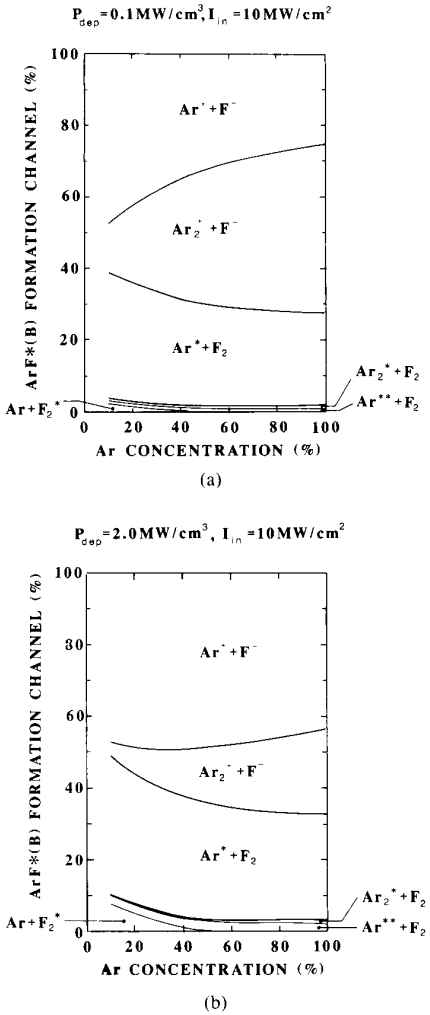


Fig. 9. Calculated ArF* formation channels as a function of Ar concentration at the low-excitation rate of (a) 0.1 MW/cm³ and high excitation rate of (b) 2.0 MW/cm³, respectively.

tion, especially through the ion–ion recombination reactions such as $Ar^+ + F^-$ and $Ar_2^+ + F^-$ at each excitation rate. This indicates that Ar-rich mixture requires less energy for producing Ar* and Ar^+, precursors for ArF*, then those in Ne-rich mixture. At the lower excitation rate, because the partial pressure of F_2 is smaller than that at the higher excitation rate, the relative number density of

Ar_2^+ increases via a charge transfer reaction listed in Table I. Then, the reaction $Ar_2^+ + F^-$ becomes the main formation channel in the Ar-rich mixture pumped at the low-excitation rate as shown in Fig. 9(a). However, at the high-excitation rate as shown in Fig. 9(b), the reaction $Ar_2^+ + F^-$ contributes less to the ArF* formation processes due to the fact that the Ar^+ ion is captured rapidly by the F^- ion to form ArF*. Also, the increase of the electron–ion recombination rate ($Ar_2^+ + e$) which increases with gas temperature competes for the Ar^+, as shown in Table I.

The Ne-rich gas mixture has a relatively large loss in energy to form the ArF* because of the formation process of NeF*. On the other hand, Fig. 10 shows the relaxation channels of ArF*(B) as a function of Ar concentration when the excitation rate is (a) 0.1 and (b) 2.0 MW/cm^3, respectively. The input laser beams are injected, at constant intensity of 10 MW/cm^2 for each mixture and each excitation rate in this study. Fractional relaxation channels are calculated by their temporal and spatial summations of the entire Gaussian pulse at the exit of the amplifier. Therefore, the fraction of the stimulated emission is seen to be relatively small due to the unsaturated at each edge of the laser pulse. The two-body and three-body quenching rates of ArF* by Ar increase with increasing Ar concentration for the two excitation rate cases, as shown in Fig. 10(a) and (b), because of the ArF* + 2Ar reaction. This reaction is dependent on the gas temperature and has a large rate constant of 5.0×10^{-31} cm^6/s [58] at room temperature. The percentage of ArF* contributing to the stimulated emission decreases with increasing Ar concentration because the competing relaxation rates of ArF* by Ar and 2Ar are faster than by Ne at the same excitation rate. Consequently, nevertheless the Ar-rich mixture forms a larger ArF* number density than does the Ar-lean mixture, the Ne-rich mixture shows a higher gain at the same excitation rate.

The nonsaturable absorption in the excimer laser amplifier restricts the amplifier gain length scaling and its extraction efficiency. Fig. 11 shows the calculated absorption coefficient α as a function of excitation rate at the peak of the small-signal gain curve. These results were obtained for conditions of optimum input intensities in the range of 5 ~ 10 MW/cm^2 for all gas mixtures, chosen to maximize the extraction efficiency. Thus, the absorption coefficient in this study is nonsaturable absorption. The absorption coefficient α is almost linearly proportional to the excitation rates for all mixtures. At the same excitation rate, differences in absorption are quite small among the three mixtures, although the absorption increases as the Ne concentration increases.

Fig. 12 shows the absorption channels at 193 nm as a function of Ar concentration under the conditions of optimum input beams for the excitation rates of (a) 0.1 and (b) 2.0 MW/cm^3, respectively. The dominant absorber is F^-, which occupies between 50–80 percent of the absorption channels at each excitation rate and for all three mixtures. Here, the F^- is mainly produced by the electron

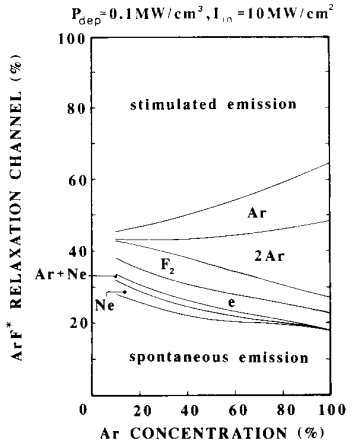


Fig. 10. Calculated relaxation channels of ArF* as a function of Ar concentration with the strong input beam intensity of 10 MW/cm^2 for all mixtures at the low-excitation rate of (a) 0.1 MW/cm^3 and high excitation rate of (b) 2.0 MW/cm^3, respectively.

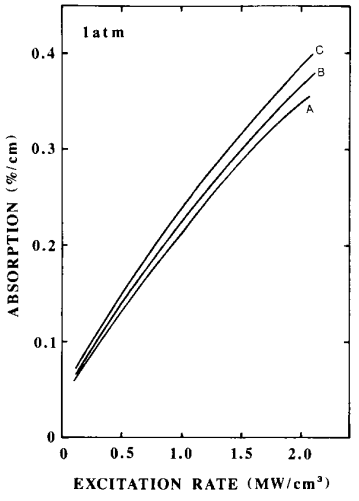


Fig. 11. Calculated absorption coefficient curves as a function of excitation rate for each optimum mixture using the optimum input intensities as shown in Fig. 6. The designation of the curves is the same as that of Fig. 6.

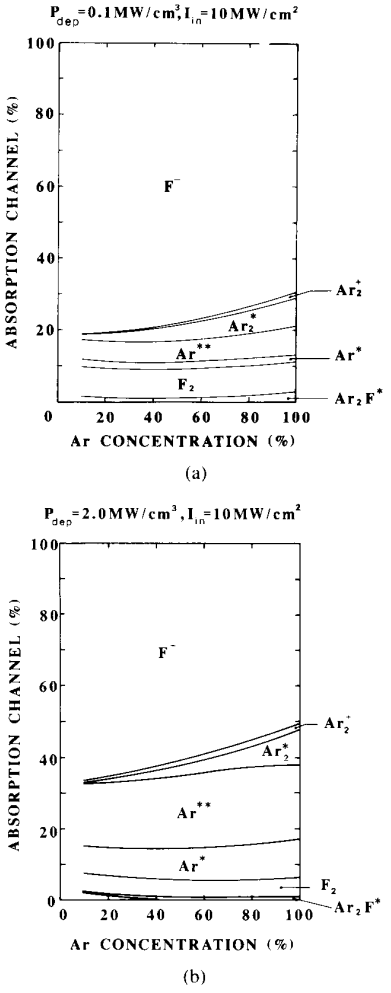


Fig. 12. Calculated fractional absorption channels as a function of Ar concentration with the existence of optimum input beam intensities at the low-excitation rate of (a) 0.1 MW/cm³ and high excitation rate of (b) 2.0 MW/cm³, respectively. The optimum input intensity used for each mixture is shown in Fig. 6.

reactions; $F_2 + e_s \rightarrow F^- + F$ and $F + e_s + M \rightarrow F^- + M$. Because the number density of the secondary low-energy electrons e_s increase with increasing the Ne-buffer concentration and the number density of F^- decrease with increasing the Ar concentration, due to the increase of ionic formation channel ($Ar^+ + F^-$ and $Ar_2^+ + F^-$), the absorption coefficient decreases with decreasing the Ar concentration.

Fig. 13 shows the gain to absorption ratio (g_0/α) as a function of excitation rate for the three mixtures. This ratio is calculated from the data shown in Fig. 9 and Fig. 12. The rate of increase of g_0 exceeds that of α with increasing excitation rate at the lower excitation rates, less than 1.0 MW/cm³. The three-body collisional quenching of the ArF* by 2Ar is reduced with increasing excitation rate due to the gas temperature rise, as shown in Fig. 10. However, the fractional contribution to the absorption by Ar_2F^* formed by the three-body quenching of the ArF* are very small even at the low-excitation rate as shown in

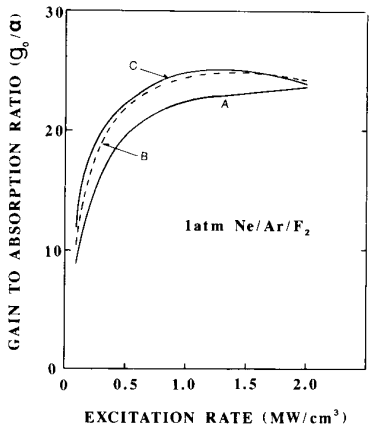


Fig. 13. Calculated gain to absorption ratio (g_0/α) as a function of excitation rate. The designation of the curves is the same as that of Fig. 6.

Fig. 12(a). This study shows that the absorption by Ar_2F^* near 285 ± 25 nm [52], [73] does not have an important effect upon absorption in the ArF laser amplifier, as compared with the role of Kr_2F^* in the KrF laser amplifier [74].

The mixtures containing the Ne-buffer gas ([Ar] = 40 and [Ar] = 70 percent) also have individual peaks of g_0/α at an excitation rate of around 1.0 MW/cm³. On the other hand, in the Ar/F_2 mixture, the g_0/α shows a slight tendency to a further increase with an increase of excitation rate over 1.0 MW/cm³ due to the decrease of the three-body quenching by 2Ar.

Fig. 14 shows the calculated saturation intensity I_s as a function of excitation rate. The Ar-richer mixtures have higher saturation intensities than the Ar-lean mixtures because the effective lifetime of the ArF* upper level becomes shorter in Ar-richer mixtures because of the fast collisional quenching of ArF* by Ar and 2Ar. The saturation intensity increases with excitation rate because of the increase in the electron quenching rate. However, in excess of 0.5 MW/cm³, the increase rate of the saturation intensity becomes less pronounced since the three-body quenching rate of ArF* decreases due to the gas temperature rise.

C. ArF Formation Efficiency, Extraction Efficiency, and Power Efficiency*

The extraction efficiency (not local) and the power efficiency are calculated based on the extracted intensities (I_{out}–I_{in}) which were obtained from the pulse propagation equation considering gain depletion and saturation of the absorber by the optimized input intensities. The ArF* formation efficiency (η_{form}), extraction efficiency (η_{ext}), and the power efficiency (η_{power}) are estimated by using the following equation:

$$\eta_{form} = \frac{h\nu R}{P_{dep}} \tag{4}$$

$$\eta_{ext} = \frac{I_{out}-I_{in}}{h\nu RL} \tag{5}$$

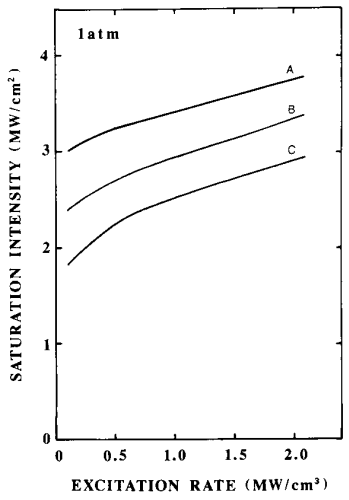


Fig. 14. Calculated saturation intensity as a function of excitation rate for each gas mixture. The designation of the curves is the same as that of Fig. 6.

$$\eta_{\text{power}} = \frac{I_{\text{out}} - I_{\text{in}}}{P_{\text{dep}} L} \tag{6}$$

where $h\nu$ is the photon energy of ArF* ($B \rightarrow X$), R is the pump rate of ArF*(B) state, and P_{dep} is the e-beam deposition power (MW/cm^3). The extraction and power efficiencies expressed in (5) and (6) considered the distribution of the laser beam intensity along the finite optical path length (L) of the pumping media [9], [75].

We used the peak value of the excitation rate for estimating the ArF* formation efficiency and the power efficiency of the ArF amplifier. Fig. 15 shows calculated ArF* formation efficiency (η_{form}) and the extraction efficiency (η_{ext}). The calculated power efficiency (η_{power}) is also shown in Fig. 16 for all mixtures studied as a function of excitation rate. As can be seen in Fig. 15 the formation efficiency (η_{form}) for all mixtures is almost constant for a wide range of excitation rates. However, the formation efficiency slightly decreases with increasing excitation rate probably because the $Ar_2^+ + F^-$ channel decreases. The Ar-rich mixtures have higher formation efficiency than those of the Ne-rich mixtures over the excitation rate range of 0.1–2.0 MW/cm^3. This means that Ar-rich mixtures can more efficiently produce the upper laser level precursors, such as Ar* and Ar^+ due to the smaller W-value of Ar than that of Ne.

The extraction efficiency of the Ar/F_2 mixture is improved by increasing the excitation rate. At low-excitation rates below 0.5 MW/cm^3, Ne-rich mixtures have higher extraction efficiencies than Ar-rich mixtures in which the three-body quenching rate is very fast. However, the extraction efficiency of each mixture tends to be nearly constant (around 70 percent) for excitation rates in excess of 1 MW/cm^3 due to increases of the electron and F_2 quenching rates which lower the fraction of stimulated emission as shown in Fig. 10.

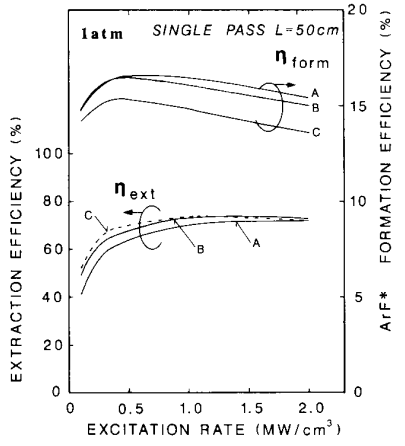


Fig. 15. Calculated ArF* formation efficiency (η_{form}) and extraction efficiency (η_{ext}) as a function of excitation rate. The designation of the curves is the same as that of Fig. 6.

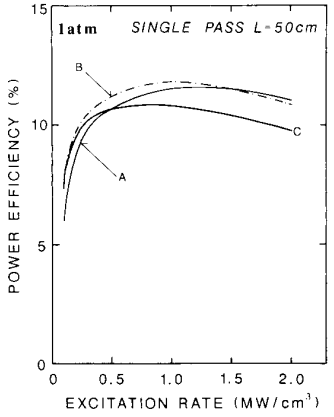


Fig. 16. Calculated power efficiency (η_{power}) as a function of excitation rate. The designation of the curves is the same as that of Fig. 6.

High-power efficiencies in excess of 10 percent are obtainable for all mixtures studied for excitation rates ranging from 0.5 to 2.0 MW/cm^3. The Ar-rich mixtures are of great advantage to the power efficiency for excitation rates in excess of 1.0 MW/cm^3. The high-power efficiency of the ArF laser amplifier over a wide range of excitation rates (0.5–2.0 MW/cm^3) obtained in this study can be compared with the result of the theoretical study of the KrF laser amplifier performed previously [74]. The power efficiencies of both amplifier systems are almost equivalent under the same range of excitation rates (0.5 ~ 2.0 MW/cm^3).

D. Stage Gain and Input Pulsewidth

Fig. 17 shows the calculated stage gain $G(I_{\text{out}}/I_{\text{in}})$ and the power efficiency η_{power} as a function of input intensity for $I_{\text{in}} = 0.1 \sim 50$ MW/cm^2 using the Ar/F_2 mixture for excitation rates of 0.5, 1.0, and 2.0 MW/cm^3. In this study, as described above, we also employ single-pass (L

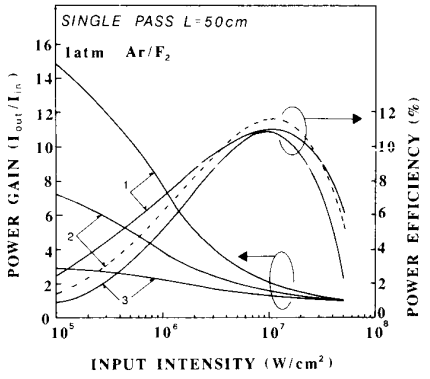


Fig. 17. Calculated stage gain $G(I_{out}/I_{in})$ and the power efficiency (η_{power}) as a function of input intensity (I_{in}) using the atmospheric mixture of Ar/F_2 for excitation rates of 0.5, 1.0, and 2.0 MW/cm^3. The definition of the curve number attached to each line is as follows. Curve 1: P_{dep} = 2.0 MW/cm^3, curve 2: P_{dep} = 1.0 MW/cm^3, and curve 3: P_{dep} = 0.5 MW/cm^3.

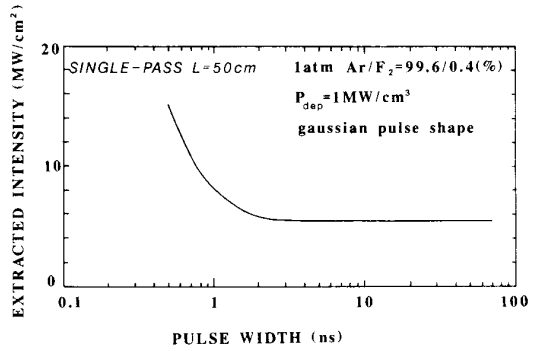


Fig. 18. Predicted variation of the extracted intensity (I_{out}-I_{in}) as a function of input beam pulsewidth ranging from 0.5 to 70 ns for the atmospheric mixture of Ar/F_2 = 99.6/0.4 (percent) pumped at the excitation rate of 1 MW/cm^3. Each pulse has a Gaussian profile and peak intensities for the various pulsewidths are optimized to get the maximum extracted intensity.

= 50 cm) amplification and use a Gaussian pulse (20 ns FWHM) for the input beam. The power gain is larger at the higher excitation rates. In the small-signal region, the rate of amplification increases significantly with an increase of excitation rate, indicating that the power gain approaches the value of $\exp(g_0 - \alpha)L$ at the high-excitation rate having a larger part of unsaturated gain than that in the low-excitation rate. At an input intensity of around 50 MW/cm^2, the value of the power gain becomes one due to the full gain saturation. The power efficiencies for the three excitation rates become maximum at an input intensity of around 10 MW/cm^3 with power gains of about 1.5 ~ 2.2.

For amplifier windows, one can use fused silica, which have a relatively larger two-photon absorption coefficient than either CaF_2 and MgF_2. Two-photon absorption is also negligible at the output intensity of 20 MW/cm^2, the value at which the power efficiency is maximized for the three values of excitation rates studied for these configurations. We assumed that the two-photon absorption coefficient (β) for fused silica at 193 nm is 5.78×10^{-11} (cm/W) converted from the value of 4.5×10^{-11} (cm/W) at 248 nm [77] scaled by the photon energy ratio of the 193 to the 248 nm. We used the approximate formula of $1/T = 1 + \beta t I_i$ [78], where T and t are the transmission and the thickness of the optical material, respectively, and I_i is the input intensity of the optical material. If the thickness of the fused silica window is 3 cm and the laser pulse length is short (<50 ns), the maximum input intensity should be 58.25 MW/cm^2 for maintaining its transmission in excess of 99.0 percent.

Fig. 18 shows a sample calculation of the extracted intensity as a function of pulsewidth. The input intensities for the various pulsewidths were also optimized to obtain the maximum extracted intensity. It is also assumed in this calculation that the input beam has a Gaussian pulse shape with various pulsewidths ranging from 0.5 to 70 ns (FWHM) at the excitation rate of 1.0 MW/cm^3. These results show that it is possible to enhance the output power of the ArF laser amplifier by means of shortening the pulse duration. As mentioned in the previous theoretical KrF amplifier study [74], if the input pulse has a leading edge as short as the collisional relaxation time of ArF* and its intensity is high enough, the leading part of the propagation pulse extracts a large portion of the energy stored in the amplifier media effectively. This results in the distortion of the amplified pulse in the leading part of the output pulse because, to some extent, the ArF laser is so-called an energy-storage laser within a very short relaxation time (<1.5 ns obtained in this study). The extracted intensities rapidly increase in the pulsewidth region less than 2 ns. In the case of the KrF laser amplifier [74], the increase of output intensity with decreasing pulsewidth begins below a pulsewidth of 10 ns. This is longer than that for ArF because of the collisional relaxation time of ArF* is shorter than that of KrF*.

IV. Conclusion

We have performed a theoretical study on the characteristics of a single-pass (L = 50 cm), atmospheric-pressure, short pulse ArF laser amplifier for a wide range of excitation rates (0.1 ~ 2.0 MW/cm^3) using pulse propagation equations which consider transient spatial and temporal dynamics. We have also reported a recommended set of kinetic rate constants for the ArF laser amplifier. The present amplifier model, which includes modified rate constants for electron quenching different for each mixture, provides good agreement with the experimental results [9]. This highlights the importance in modeling rare gas halide lasers of determining the precise rate constant for electron quenching appropriate for various mixtures. More study is definitely necessary to understand secondary electron physics and F_2 destruction mechanisms in order to obtain the exact rate constant for electron quenching.

It is found that the absorption of Ar_2F^* does not play an important role in the ArF laser amplifier, even at low-excitation rates, in contrast to the significant effect of Kr_2F^* in the KrF laser amplifier.

As a result of this model, ArF laser power amplifier efficiencies of over 10 percent are predicted for Ar-rich mixtures at excitation rates ranging from 0.5 to 2.0 MW/cm^3, which is comparable to that of KrF [74]. It is also determined that the ArF laser has smaller absorption and larger saturation intensities than does the KrF laser for the same excitation rate [74].

Finally, although nonlinear optical problems such as two-photon absorption in fused silica window materials at 193 nm for an ArF laser system at intensities of $\sim GW/cm^2$ level may emerge, the ArF amplifier is found to operate efficiently at intensities of 20 MW/cm^2, at which such absorption will not be crucial. If one wants to operate an ArF amplifier at the $\sim GW/cm^2$ power level, multiphoton absorption can be reduced by selecting optical materials like CaF_2 and MgF_2, which have larger band gaps than that of fused silica. This strongly suggests that the ArF laser can be an alternative to the KrF laser as an ICF energy driver.

Acknowledgment

The authors would like to thank Dr. A. Suda for his contribution to the development of the basic ArF laser oscillator simulation code and for his fruitful suggestions on this subject. They also wish to thank H. Kumagai for his helpful discussions during this work.

References

[1] H. Hemmati and G. J.Collins, "Atomic Sn, Sb, Ge photodissociation lasers," *IEEE J. Quantum Electron.*, vol. QE-16, pp. 1014-1016, Oct. 1980.

[2] A. W. McCown and J. G. Eden, "ZnI($B \rightarrow X$) laser: 600-604 nm," *Appl. Phys. Lett.*, vol. 40, pp. 371-373, Sept. 1981.

[3] T. F. Deutsch, J. C. C. Fan, G. W. Turner, R. L. Chapman, D. J. Ehrilich, and R. M. Osgood, Jr., "Efficient Si solar cells by laser photochemical doping," *Appl. Phys. Lett.*, vol. 38, pp. 144-146, Feb. 1981.

[4] T. F. Deutsch, D. J. Ehrlich, D. D. Rathman, D. J. Silversmith, and R. M. Osgood, Jr., "Electrical properties of laser chemically doped silicon," *Appl. Phys. Lett.*, vol. 39, pp. 825-827, Nov. 1981.

[5] R. W. Andreatta, C. C. Abele, J. F. Osmundsen, and J. G. Eden, "Low-temperature growth of polycrystalline Si and Ge films by ultraviolet laser photodissociation of silane and germane," *Appl. Phys. Lett.*, vol. 40, pp. 183-185, Jan. 1982.

[6] D. H. Lowndes, D. B. Geohegan, D. Eres, S. J. Pennycook, D. N. Mashburn, and G. E. Jellison, Jr., "Photon-controlled fabrication of amorphous superlattice structures using ArF(193 nm) excimer laser," *Appl. Phys. Lett.*, vol. 52, pp. 1868-1870, May 1988.

[7] F. K. Cross, R. K. Al-Dhahir, P. E. Dyer, U. Hull, and R. Srinivasan, "Photoacoustic studies of UV laser tissue ablation," in *Tech Dig. Conf. Lasers, Electro-Opt.*, vol. 7, pp. 312-314, Apr. 1988.

[8] W. L. Kruer, *Progress in Laser and Laser Fusion*. New York: Plenum, 1975, p. 5.

[9] Y. W. Lee, H. Kumagai, S. Ashidate, and M. Obara, "High-efficiency extraction study of an electron beam pumped ArF laser amplifier with an atmospheric-pressure Ar-rich mixture," *Appl. Phys. Lett.*, vol. 52, pp. 1294-1296, Apr. 1988.

[10] A. Suda, M. Obara, and A. Noguchi, "Performance characteristics of the ArF excimer laser using a low-pressure argon-rich mixture," *J. Appl. Phys.*, vol. 60, pp. 3791-3793, Nov. 1986.

[11] A. Suda, H. Kumagai, S. Izawa, and M. Obara, "Comparative study of low-pressure rare-gas fluoride/chloride lasers excited by a short-pulse electron beam," *J. Appl. Phys.*, vol. 64, pp. 1720-1725, Aug. 1988.

[12] A. Mandl, "ArF short pulse extraction studies," *J. Appl. Phys.*, vol. 59, pp. 1435-1445, Mar. 1986.

[13] A. Suda, M. Obara, and T. Fujioka, "Kinetic studies of the electron-beam-pumped ArF laser using an $Ne/Ar/F_2$ mixture," *Japan. J. Appl. Phys.*, vol. 24, pp. 1183-1188, Sept. 1985.

[14] J. M. Hoffman, A. K. Hays, and G. C. Tisone, "High-power uv noble-halide lasers," *Appl. Phys. Lett.*, vol. 28, pp. 538-539, May 1976.

[15] J. R. Murray and H. T. Powell, *Electron Transition Lasers II*. Cambridge, MA: M.I.T., 1977, p. 23.

[16] C. B. Edwards, F. O'Neill, and M. J. Shaw, "60 ns e-beam excitation of rare-gas halide lasers," *Appl. Phys. Lett.*, vol. 36, pp. 617-620, Apr. 1980.

[17] P. J. M. Peters, I. H. T. Fierkens, and W. J. Witteman, "Effect of Ne and Ar on the performance of a high-pressure ArF* laser pumped by a small coaxial electron beam," *Appl. Phys. Lett.*, vol. 51, pp. 883-885, Sept. 1987.

[18] A. D. Klementov, N. V. Morozov, and P. B. Sergeev, "Electron-beam-excited ArF laser," *Sov. J. Quantum Electron.*, vol. 16, pp. 1139-1142, Aug. 1986.

[19] F. Kannari, M. Obara, and T. Fujioka, "An advanced kinetic model of electron-beam-excited KrF lasers including the vibrational relaxation in KrF*(B) and collisional mixing of KrF*(B, C)," *J. Appl. Phys.*, vol. 57, pp. 4309-4322, May 1985.

[20] A. Suda, H. Kumagai, and M. Obara, "Characteristics of an electron beam pumped KrF laser amplifier with an atmospheric-pressure Kr-rich mixture in a strongly saturated region," *Appl. Phys. Lett.*, vol. 51, pp. 218-220, July 1987.

[21] D. D. Lowenthal and J. M. Eggleston, "ASE effects in small aspect ratio laser oscillators and amplifiers with nonsaturable absorption," *IEEE J. Quantum Electron.*, vol. QE-22, pp. 1165-1173, Aug. 1986.

[22] A. M. Hunter, II and R. O. Hunter, Jr., "Bidirectional amplification with nonsaturable absorption and amplified spontaneous emission," *IEEE J. Quantum Electron.*, vol. QE-17, pp. 1879-1887, Sept. 1981.

[23] A. M. Hunter, II, R. O. Hunter, Jr., and T. H. Johnson, "Scaling of KrF lasers for inertial confinement fusion," *IEEE J. Quantum Electron.*, vol. QE-22, pp. 386-404, Mar. 1986.

[24] F. Kannari and W. D. Kimura, "Transient dynamics of multiplexed short-pulse amplification in KrF," *J. Appl. Phys.*, vol. 63, pp. 1299-1305, Mar. 1988.

[25] D. C. Lorents, "The physics of electron beam excited rare gases at high densities," *Physica*, vol. 82C, pp. 19-26, Mar. 1976.

[26] L. G. Christophorou, *Atomic and Molecular Radiation Physics*. New York: Wiley, 1971.

[27] A. V. Hazi, T. N. Rescigno, and A. E. Orel, "Theoretical study of the deexcitation of KrF and XeF excimers by low-energy electrons," *Appl. Phys. Lett.*, vol. 35, pp. 477-479, Oct. 1979.

[28] L. A. Levin, S. E. Moody, E. L. Klosterman, R. E. Center, and J. J. Ewing, "Kinetic model for long-pulse XeCl laser performance," *IEEE J. Quantum Electron.*, vol. QE-17, pp. 2282-2289, Dec. 1981.

[29] "Conceptual design of a KrF laser scaling module," MSNW, Inc., Perkin-Elmer, and Maxwell Labs., Inc., Final Rep., U.S. Dep. Energy, issued Oct. 1980 (Contract DE-AC08-79DP40115), unpublished.

[30] D. W. Trainor and J. H. Jacob, "Electron dissociative attachment rate constants for F_2 and NF_3 at 300 and 500°K," *Appl. Phys. Lett.*, vol. 35, pp. 920-922, Dec. 1979.

[31] T. H. Johnson and A. M. Hunter, II, "Physics of the krypton fluoride laser," *J. Appl. Phys.*, vol. 51, pp. 2406-2420, May 1980.

[32] D. L. Huestis, R. M. Hill, H. H. Nakano, and D. C. Lorents, "Quenching of Ne*, F*, and F_2* in $Ne/Xe/NF_3$ and $Ne/Xe/F_2$ mixtures," *J. Chem. Phys.*, vol. 69, pp. 5133-5139, Dec. 1978.

[33] E. Ellis and N. D. Twiddy, "Time-resolved optical absorption measurements of excited-atom concentrations in the argon afterglow," *J. Phys. B*, vol. 2, pp. 1366-1377, Dec. 1969.

[34] Analogous to $Ar^* + 2Ar \rightarrow Ar_2^* + Ar$, E. Ellis and N. D. Twiddy, "Time-resolved optical absorption measurements of excited-atom concentrations in the argon afterglow," *J. Phys. B*, vol. 2, pp. 1366-1377, Dec. 1969.

[35] Estimated from [28] and references therein.

[36] Estimated from [29] and references therein.

[37] Analogous to $Ar^{**} + Ar \rightarrow Ar^* + Ar$, "Conceptual design of a KrF

laser scaling module," MSNW, Inc., Perkin-Elmer, and Maxwell Labs., Inc., Contract DE-AC08-79DP40115, Oct. 1980, and references therein.

[38] Analogous to $Ne^* + Xe \rightarrow Xe^+ + Ne + e$, R. H. Neynaber and S. Y. Tang, "Chemi-ionization in collisions of metastable neon with xenon," *J. Chem. Phys.*, vol. 70, pp. 4271–4276, May 1979; S. Y. Tang, A. B. Marcus, and E. E. Muschlits, Jr., "Velocity dependence of the ionization of Ar, Kr, and Xe on impact of metastable neon atoms," *J. Chem. Phys.*, vol. 56, pp. 566–572, Jan. 1972.

[39] A. P. Vitols and H. J. Oskam, "Reaction rate constant for $Ne^+ + 2Ne \rightarrow Ne_2^+ + Ne$," *Phys. Rev. A*, vol. 5, pp. 2618–2622, June 1972.

[40] Estimated from other heterodimer-ion formation: $Ne^+ + Xe + Ne \rightarrow NeXe^+ + Ne$, T. G. Finn, L. J. Palumbo, and L. F. Champagne, "A kinetics scheme for the XeF laser," *Appl. Phys. Lett.*, vol. 33, pp. 148–151, July 1978; and $Ar^+ + Kr + Ne \rightarrow ArKr^+ + Ar$, W. B. Lacina and D. B. Cohn, "Theoretical anlaysis of the electrically excited KrF laser," *Appl. Phys. Lett.*, vol. 32, pp. 106–108, Jan. 1978.

[41] D. Smith and P. R. Cromey, "Conversion rates and ion mobilities in pure neon and argon afterglow plasmas," *J. Phys. B*, vol. 1, pp. 638–649, July 1968.

[42] H. J. Oskam, "Microwave investigation of disintegrating gaseous discharge plasmas," *Phillips Res. Rep.*, vol. 13, pp. 401–457, Oct. 1958.

[43] C. B. Collins and F. W. Lee, "Measurement of the rate coefficients for bimolecular and termolecular ion-molecule reactions of Ne_2^+ with selected atomic and molecular species," *J. Chem. Phys.*, vol. 72, pp. 5381–5389, May 1980.

[44] H. L. Kramer, J. A. Herce, and E. E. Muschlitz, Jr., "Formation and collisional dissociation of heteronuclear rare-gas associative ions," *J. Chem. Phys.*, vol. 56, pp. 4166–4170, Apr. 1972.

[45] J. N. Bardsley, and M. A. Biondi, *Advances in Atomic and Molecular Physics*, D. R. Bates and J. Esteman, Eds. New York: Academic, 1970.

[46] Analogous to $Ar_2^+ + e \rightarrow Ar^{**} + Ar$, J. N. Bardsley and M. A. Biondi, Advances in *Atomic and Molecular Physics*, D. R. Bates and J. Esteman, Eds. New York: Academic, 1970.

[47] B. Schneider and J. S. Cohen, "Ground and excited states of Ne_2 and Ne_2^+.II. Spectroscopic properties and radiative lifetimes," *J. Chem. Phys.*, vol. 61, pp. 3240–3243, Oct. 1974.

[48] J. W. Keto, R. E. Gleason, Jr., and G. K. Walters, "Production mechanisms and radiative lifetimes of argon and xenon molecules emitting in the ultraviolet," *Phys. Rev. Lett.*, vol. 33, pp. 1365–1368, Dec. 1974.

[49] J. M. Hoffman and J. B. Moreno, Sandia Nat. Labs., Rep. SAND80-1486, Albuquerque, NM, 1980.

[50] J. E. Velazco, J. H. Koltz, and D. W. Setser, "Quenching rate constants for metastable argon, krypton, and xenon atoms by fluorine containing molecules and branching ratios for XeF* and KrF* formation," *J. Chem. Phys.*, vol. 65, pp. 3468–3480, Nov. 1976.

[51] C. H. Chen, M. G. Payne, and J. P. Judish, "Kinetic studies of ArF* and Ar_2F^* in proton-excited Ar–F_2 mixtures," *J. Chem. Phys.*, vol. 69, pp. 1625–1635, Aug. 1978.

[52] Some of the Ar_2^* quenched by the F_2 seems to produce Ar_2F^*, G. Marowsky, G. P. Glass, F. K. Tittel, K. Hohla, W. L. Willson, Jr., and H. Weber, "Formation kinetics of the triatomic excimer Ar_2F^*," *IEEE J. Quantum Electron.*, vol. QE-18, pp. 898–902, May 1982. The rate constant is assumed to be analogous to $Ar_2^* + NF_3 \rightarrow Ar_2F^* + NF_2$. N. Boewering, R. Sauerbrey, and H. Langhoff, "Kinetic studies of Ar_2F^* in electron-beam excited Ar-NF_3 mixtures," *J. Chem. Phys.*, vol. 76, pp. 3524–3528, Apr. 1982.

[53] Analogous to $Ar^* + F_2 \rightarrow ArF^* + F$, J. E. Velazco, J. H. Koltz, and D. W. Setser, "Quenching rate constants for metastable argon, krypton, and xenon atoms by fluorine containing molecules and branching ratios for XeF* and KrF* formation," *J. Chem. Phys.*, vol. 65, pp. 3468–3480, Nov. 1976.

[54] Analogous to $Ne^* + F_2 \rightarrow NeF^* + F$, D. L. Huestis, R. M. Hill, H. H. Nakano, and D. C. Lorents, "Quenching of Ne*, F*, and F_2* in Ne/Xe/NF_3 and Ne/Xe/F_2 mixtures," *J. Chem. Phys.*, vol. 69, pp. 5133–5139, Dec. 1978.

[55] R. F. S. Chang, "Electron-beam pumped kinetic study of ArF*(B) state," *J. Chem. Phys.*, vol. 78, pp. 2210–2213, Mar. 1983.

[56] M. Rokni, J. H. Jacob, J. A. Mangano, and R. Brochu, "Formation and quenching kinetics of ArF*," *Appl. Phys. Lett.*, vol. 31, pp. 79–82, July 1977.

[57] D. W. Trainor and J. H. Jacob, "Electron quenching of KrF* and ArF*," *Appl. Phys. Lett.*, vol. 37, pp. 675–677, Oct. 1980.

[58] N. Boewering, R. Sauerbrey, and H. Langhoff, "Kinetic studies of Ar_2F^* in electron-beam excited Ar-NF_3 mixtures," *J. Chem. Phys.*, vol. 76, pp. 3524–3528, Apr. 1982.

[59] Estimated from $ArF^* + 2Ar \rightarrow Ar_2F^* + Ar$ and $ArF^* + He + Ar \rightarrow Ar_2F^* + He$, N. Boewering, R. Sauerbrey, and H. Langhoff, "Kinetic studies of Ar_2F^* in electron-beam excited Ar-NF_3 mixtures," *J. Chem. Phys.*, vol. 76, pp. 3524–3528, Apr. 1982.

[60] Based on the radiative lifetime, calculated by T. H. Dunning, Jr. and P. J. Hay, "The covalent and ionic states of the rare gas monofluorides," *J. Chem. Phys.*, vol. 69, pp. 134–149, July 1978. The bandwidth was measured by Yu. A. Kudryavtsev and N. P. Kuzmina, "Excimer gas-discharge tunable ArF laser," *Appl. Phys.*, vol. 13, pp. 107–108, May 1977.

[61] G. Marowsky, G. P. Glass, F. K. Tittel, K. Hohla, W. L. Wilson, Jr., and H. Waber, "Formation kinetics of the triatomic excimer Ar_2F^*," *IEEE J. Quantum Electron.*, vol. QE-18, pp. 898–902, May 1982.

[62] Estimated from comparison of calculated and measured NeF* fluorescence efficiency in Ne/F_2 mixture. The experiments were done by J. K. Rice, J. W. Woodworth, and A. K. Hays, Sandia Nat. Labs., Rep. SAND77-1517, Albuquerque, NM, 1978.

[63] H. H. Michels, R. H. Hobbs, and L. A. Wright, "Electron structure of the noble gas dimer ions.II. Theoretical absorption spectrum for the $A^2_{1/2u} \rightarrow D^2_{1/2g}$ system," *J. Chem. Phys.*, vol. 71, pp. 5053–5062, Dec. 1979.

[64] R. Sauerbrey, H. Eizenhofer, U. Schallar, and H. Langhoff, "Photoexcitation of the rare-gas excimer molecules Ne_2^*, Ar_2^*, and Kr_2^*," *J. Phys. B*, vol. 19, pp. 2279–2289, Aug. 1986.

[65] C. Duzy, J. C. Hsia, H. Hyman, J. H. Jacob, D. Klimek, H. Parks, and D. Trainor, "Rare gas halide lasers for fusion," Final Tech. Rep. NTIS HC A06/MF A01, Avco Everett Res. Lab., Inc., Everett, MA, Apr. 1980.

[66] R. J. Robinson and S. Geltman, "Single- and double-quantum photodetachment of negative ions," *Phys. Rev.*, vol. 153, pp. 4–8, Jan. 1967.

[67] Estimated from absorption spectrum over 2100 A, R. K. Steunenberg, and R. C. Vogel, "The absorption spectrum of fluorine," *J. Amer. Chem. Soc.*, vol. 78, pp. 901–902, Mar. 1956.

[68] C. J. Elliot and A. E. Greene, "Electron energy distributions in e-beam generated Xe and Ar plasmas," *J. Appl. Phys.*, vol. 47, pp. 2946–2953, July 1976.

[69] Z. Rozenberg, M. Lando, and M. Rokni, "Direct measurement of the electron density in electron-beam-irradiated Ar–F_2 gas mixtures by time-resolved interferometry," *Phys. Rev. D*, vol. 35, pp. 4151–4159, May 1987.

[70] F. Kannari and W. D. Kimura, "Low-energy electron distribution in electron-beam-excited XeCl laser mixture," *J. Appl. Phys.*, vol. 64, pp. 500–506, July 1988.

[71] H.-L. Chen, R. E. Center, D. W. Trainor, and W. I. Fyfe, "Dissociative attachment of electrons to F_2," *Appl. Phys. Lett.*, vol. 30, pp. 99–101, Jan. 1977.

[72] M. J. Kushner and T. J. Moratz, "Direct dissociation of F_2 in electron beam pumped excimer lasers: The effect of electron density," *Appl. Phys. Lett.*, vol. 52, pp. 1856–1858, May 1988.

[73] D. C. Lorents, D. L. Huestis, M. V. McCusker, H. H. Nakano, and R. M. Hill, "Optical emissions of triatomic rare gas halides," *J. Chem. Phys.*, vol. 68, pp. 4657–4661, 1978.

[74] Y. W. Lee, F. Kannari, and M. Obara, "Power extraction study of an e-beam pumped atmospheric pressure, Kr-rich KrF laser amplifier," *J. Appl. Phys.*, to be published.

[75] J. K. Rice, G. C. Tisone, and E. L. Patterson, "Oscillator performance and energy extraction from a KrF laser pumped by a high-intensity relativistic electron beam," *IEEE J. Quantum Electron.*, vol. QE-16, pp. 1315–1326, Dec. 1980.

[76] T. Izawa, Y. Ishiwata, I. Hashimoto, H. Shikakura, Y. Owadano, Y. Matsumoto, and M. Yano, "Laser damage resistance of reflector for ArF laser," *Extended Abstracts* (35th Spring Meet., 1988), *Japan Soc. Appl. Phys., Related Soc.*, no. 3, p. 784.

[77] A. J. Taylor, R. B. Gibson, and J. P. Roberts, "Two-photon absorption at 248 nm in ultraviolet window materials," *Opt. Lett.*, vol. 13, pp. 814–816, Oct. 1988.

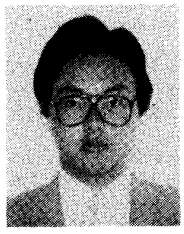
[78] S. Watanabe, A. Endoh, M. Watanabe, and N. Sarukura, "Tera watt class excimer lasers," *Oyo Buturi* (publication of Japan Soc. Appl. Phys.), vol. 57, pp. 1528–1534, Oct. 1988.



Young-Woo Lee was born in Seoul, Korea, on April 29, 1959. He received the B.S. and M.S. degrees in electronic engineering from Kyung-Hee University, Seoul, Korea, in 1983 and 1985, respectively.

During the period 1985–1986 he was drafted into the Korean Army. Since 1987 he has been working towards the Ph.D. degree in electrical engineering at the graduate school of Keio University, Yokohama, Japan. His current research interests include theoretical and experimental study of electron-beam-pumped rare-gas-halide excimer (ArF/KrF) lasers and short wavelength (VUV) gas lasers.

Mr. Lee is a member of the Japan Society of Applied Physics, the Korean Institute of Electrical Engineers, and the Korean Institute of Telecommunication and Electronics.



Eiichi Matsui was born in Tokyo, Japan, on August 19, 1965. He received the B.S. degree from Keio University, Yokohama, Japan, in 1988.

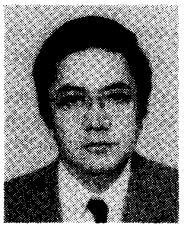
He is currently a graduate student at Keio University. His main interest is in the field of high-power excimer (rare-gas-halide) lasers and pulsed rare-gas atomic transition lasers.



Fumihiko Kannari was born in Japan on January 15, 1958. He received the B.S., M.S., and Ph.D. degrees from Keio University, Yokohama, Japan, in 1980, 1982, and 1985, respectively, all in electrical engineering.

From 1984, he was a postdoctoral research associate at SERC, Rutherford Appleton Laboratory, Oxon, England, and was involved in the development of a "Sprite" multiplexed shortpulse KrF laser system. In 1986 he joined Spectra Technology, Inc., and performed various theoretical and experimental studies on gas laser kinetics, especially rare-gas-halide excimer lasers. In 1988 he joined the Department of Electrical Engineering, Keio University, where he is currently an Assistant Professor. His main interests are currently in the field of high-power short wavelength gas lasers and their kinetics, and nonlinear pulse propagation.

Dr. Kannari is a member of the Optical Society of America, the Japan Society of Applied Physics, the Institute of Electrical Engineers of Japan, and the Laser Society of Japan.



Minoru Obara (M'82) was born in Tokyo, Japan, on September 29, 1947. He received the B.S., M.S., and Ph.D. degrees from Keio University, Yokohama, Japan, in 1971, 1973, and 1976, respectively.

In 1976 he joined the faculty at Keio University, where he is currently an Associate Professor. He has been engaged in research on high-power pulsed molecular gas lasers such as HF/DF, CO_2, rare-gas-halide excimers, VUV/XUV lasers, and pulsed-power technology such as optically-controlled diffused discharge opening switches and magnetic pulse compressors.

Dr. Obara is a member of the Institute of Electronics and Communication Engineers of Japan, the Japan Society of Applied Physics, the Laser Society of Japan, the Institute of Electrical Engineers of Japan, APS, SPIE, and OSA.